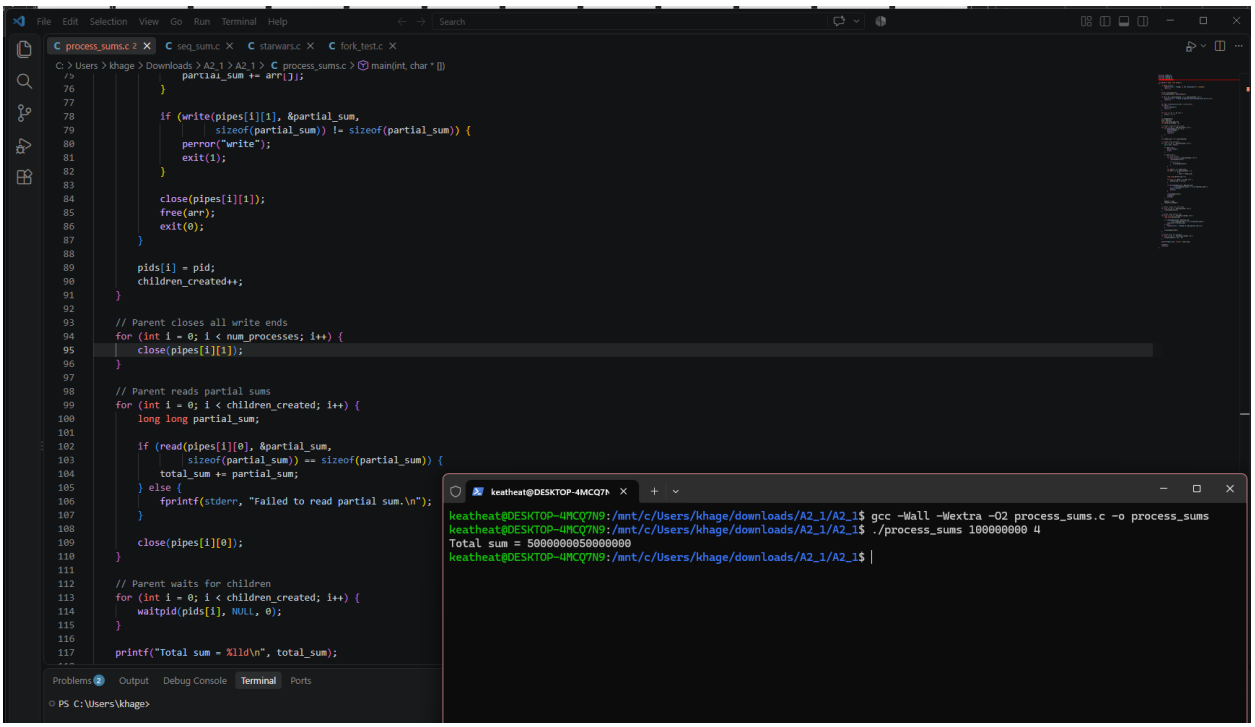
1. When the parent process creates a child using fork, the child will get its own separate copy of the parent's memory. Any changes made by the child do not affect the parent, so that value remains at 50%

Program	Processes	Time Taken (seconds)
Sequential	1	0.296
Process	1	0.379
Process	2	0.353
Process	3	0.316
Process	4	0.331

2. It is possible for the parent and child processes to run at different times, so the print statements are not guaranteed to happen in a specific order

The programs were testing using an array of 100,000,000 integers. Parallel was tested using 1,2,3 and 4 procersses. Each produced an expected sum of



5,000,000,050,000,000. Sequential was the fastest of all the tests, finishing in .296 seconds. For the process, 3 was the fastest at .316 seconds.

These results show that more processes does not constistently improve performance. The times of 2 processess and 4 processes implies that creating and managing multiple processes can add overhead that will reduce the benefits of splitting the work.

Through these tests we can see that parallel processing does not automatically make a program faster.